```
===== MODIFIED GRAPH OPERATIONS =====
1. Insert Vertex
2. Insert Edge
3. Delete Vertex
4. Delete Edge
5. Display Adjacency Matrix
6. Display Adjacency List
7. BFS Traversal
8. DFS Traversal
9. Update Graph
10. Check Path Connectivity
11. Compare Traversal
12. Exit
Enter choice: 1
Vertex 3 added.

===== MODIFIED GRAPH OPERATIONS =====
1. Insert Vertex
2. Insert Edge
3. Delete Vertex
4. Delete Edge
5. Display Adjacency Matrix
6. Display Adjacency List
7. BFS Traversal
8. DFS Traversal
9. Update Graph
10. Check Path Connectivity
11. Compare Traversal
12. Exit
Enter choice: 2
Enter two vertex numbers: 0 1
Edge added successfully.

===== MODIFIED GRAPH OPERATIONS =====
1. Insert Vertex
2. Insert Edge
3. Delete Vertex
4. Delete Edge
5. Display Adjacency Matrix
6. Display Adjacency List
7. BFS Traversal
8. DFS Traversal
9. Update Graph
10. Check Path Connectivity
11. Compare Traversal
12. Exit
Enter choice: 2
Enter two vertex numbers: 1 2
Edge added successfully.
```

```
===== MODIFIED GRAPH OPERATIONS =====
1. Insert Vertex
2. Insert Edge
3. Delete Vertex
4. Delete Edge
5. Display Adjacency Matrix
6. Display Adjacency List
7. BFS Traversal
8. DFS Traversal
9. Update Graph
10. Check Path Connectivity
11. Compare Traversal
12. Exit
Enter choice: 6

Adjacency List:
0 -> 1
1 -> 0 2
2 -> 1
3 ->

===== MODIFIED GRAPH OPERATIONS =====
1. Insert Vertex
2. Insert Edge
3. Delete Vertex
4. Delete Edge
5. Display Adjacency Matrix
6. Display Adjacency List
7. BFS Traversal
8. DFS Traversal
9. Update Graph
10. Check Path Connectivity
11. Compare Traversal
12. Exit
Enter choice: 7
Enter starting vertex: 8
Invalid starting vertex.

===== MODIFIED GRAPH OPERATIONS =====
1. Insert Vertex
2. Insert Edge
3. Delete Vertex
4. Delete Edge
5. Display Adjacency Matrix
6. Display Adjacency List
7. BFS Traversal
8. DFS Traversal
9. Update Graph
10. Check Path Connectivity
11. Compare Traversal
12. Exit
Enter choice: 10
Enter source and destination: 0 3
Path unavailable: vertices are disconnected.
```

```
===== MODIFIED GRAPH OPERATIONS =====
1. Insert Vertex
2. Insert Edge
3. Delete Vertex
4. Delete Edge
5. Display Adjacency Matrix
6. Display Adjacency List
7. BFS Traversal
8. DFS Traversal
9. Update Graph
10. Check Path Connectivity
11. Compare Traversal
12. Exit
Enter choice: 9

1. Insert Vertex
2. Insert Edge
3. Delete Vertex
4. Delete Edge
Enter update choice: 2
Enter two vertex numbers: 2 3
Edge added successfully.

Graph after update:

Adjacency List:
0 -> 1
1 -> 0 2
2 -> 1 3
3 -> 2

Previous graph had 4 vertices; updated graph has 4 vertices.
Traversal comparison: run BFS/DFS before and after the update to compare the results.

===== MODIFIED GRAPH OPERATIONS =====
1. Insert Vertex
2. Insert Edge
3. Delete Vertex
4. Delete Edge
5. Display Adjacency Matrix
6. Display Adjacency List
7. BFS Traversal
8. DFS Traversal
9. Update Graph
10. Check Path Connectivity
11. Compare Traversal
12. Exit
Enter choice: 10
Enter source and destination: 0 3
Path is available.
```